

Do You Get What You Pay For on Steam? Relating game price to the share of positive user reviews

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Introduction

Steam is one of the largest online stores for PC games. Players can buy, download, and play games on this platform, and most importantly, after playing they can leave a review: a simple thumbs up (positive) or thumbs down (negative). Games on Steam range from free to well over $60, so a possibly interesting question for both players and game developers is whether paying more also influences the number of positive reviews relative to the number of negative reviews.

In this report, we address the following question: **How does the price of a game on Steam relate to player satisfaction, measured as the proportion of positive user reviews, and is there a price range in which satisfaction is highest?**

Data sample

The dataset [1] is a public Kaggle dataset that is updated daily with information from the Steam store. The file used, `steam_games.csv`, contains 160,873 rows and 42 columns. The relevant columns for this analysis are `app_id` (unique game identifier), `name`, `price` (current price in USD) , and `positive`/`negative` (number of positive and negative user reviews).

The data was cleaned in three steps. First, 18,441 duplicate rows (the same `app_id` listed twice) were removed, leaving 142,432 unique games. Then 1,293 games without a known price and 19 games priced above $200 (treated as data errors) were removed. Finally, only games with at least 10 reviews were kept. For a game with very few reviews, the percentage of positive reviews is unreliable: a game with only one positive review would score 100%, and a single extra negative review would drop it to 50%. This leaves 55,894 games, which together still contain about 144 million reviews (96% of all reviews in the data). For each game, the satisfaction is defined as

$$S = \frac{N_{\text{positive}}}{N_{\text{positive}} + N_{\text{negative}}}.$$

Results and discussion

The games were divided into eight price ranges. Figure 1 shows how many games fall in each range: Most games on Steam fall within the lower bins, with 72% of the games costing less than $10 and only 820 games (1.5%) costing $40 or more.

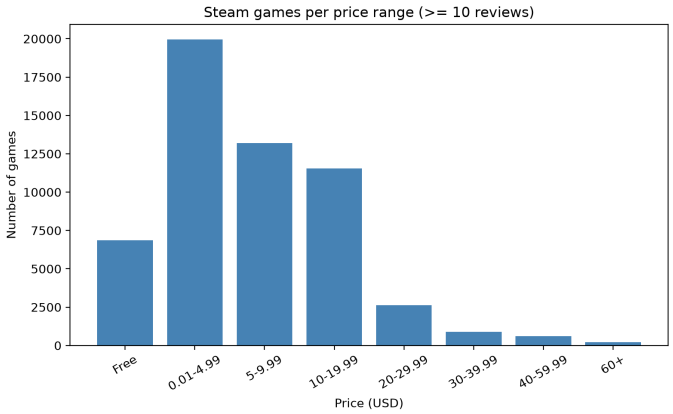


Figure 1: Number of Steam games per price range, for the 55,894 games with a known price and at least 10 user reviews.

For every price range, the mean satisfaction was calculated, with the standard error of the mean, $\sigma = s/\sqrt{N}$, as its uncertainty [2], where s is the standard deviation and N the number of games. The median was also calculated, as it is less sensitive to extreme values. The result is shown in Figure 2.

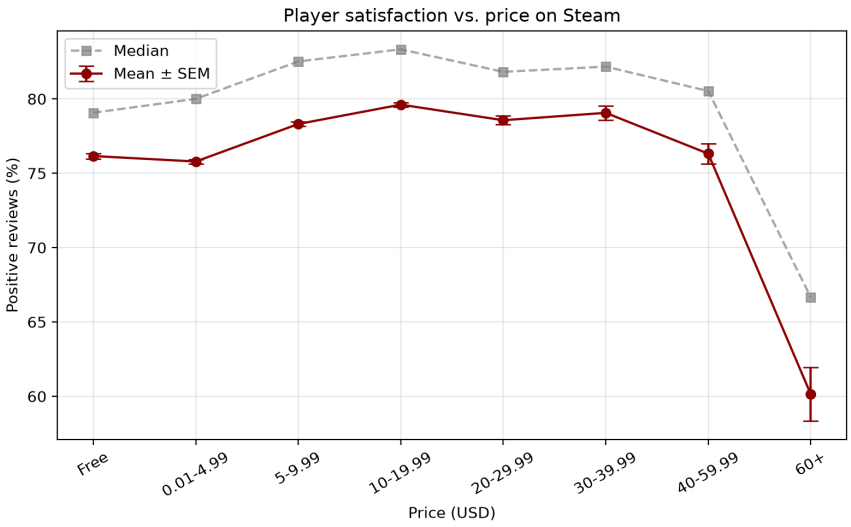


Figure 2: Player satisfaction (percentage of positive user reviews) per price range. Red points show the mean with the standard error of the mean (SEM) as error bars; grey squares show the median.

Satisfaction is highest for games costing \$10–19.99, with $S = 79.6 \pm 0.2\%$ positive reviews (11,554 games), and lowest for games of \$60 and more, with $S = 60.1 \pm 1.8\%$ (208 games). The median follows the same pattern (83.3% and 66.7%), so the result is not caused by a few extreme games. Assuming the price ranges are independent, the uncertainty of a difference is $\sigma_{\text{diff}} = \sqrt{\sigma_1^2 + \sigma_2^2}$. This gives

$$\Delta S = S_{\$10-19.99} - S_{\$60+} = 19.5 \pm 1.8 \text{ percentage points},$$

a difference of about 11σ. The \$10–19.99 range is also higher than free games (3.5 ± 0.25 points, 14σ) and games below \$5 (3.8 ± 0.20 points, 19σ).

The \$60+ range is a special case: it contains a few very popular big-budget games, but mostly obscure games with few reviews (median of only 32 reviews). When all reviews in this range are added together, 71.5% of them are positive, much higher than the average per game of 60.1%. This shows that the many obscure games with low ratings pull the average down.

This analysis shows a correlation, not causation: a higher or lower price does not by itself make players more satisfied. Moreover, the differences between price ranges are small (a few percentage points for most ranges) compared to the differences between games within the same range (standard deviation of about 18 percentage points), so price is only one of many factors that determine how well a game is received. In addition, the price is the current price (not the launch price, and discounts are ignored), and only players who chose to leave a review are counted. A next step would be to use stricter cutoffs for the cleanup and control for other factors, such as total playtime, as a higher total playtime could also be linked to a higher satisfaction among its users.

In conclusion, there seems to be a **sweet spot at \$10–19.99**, where players are most satisfied. Whereas games of \$60 and more receive the lowest share of positive reviews, although this does seem to be affected by the amount of obscure \$60+ games with very few reviews.

References

[1] H. Sidorowicz. *Steam Games Dataset (daily updates)*. Kaggle, 2026. Available at: https://www.kaggle.com/datasets/hubertsidorowicz/steam-games-dataset-daily-updates.

[2] G. Cumming, F. Fidler and D. L. Vaux. Error bars in experimental biology. *The Journal of Cell Biology*, 177(1):7–11, 2007. doi:10.1083/jcb.200611141.